

Link:

<https://github.com/farrukhshahid25/LibraryBookIssueSystem.git>

List of Implemented Exceptions:

Custom Exceptions:

1. **EmptyFieldException:** Thrown when any required text field (like Student Name, Roll Number, Book Title, or Dates) is left blank.
2. **InvalidRollNumberException:** Thrown when the user inputs letters or special characters in the Roll Number field instead of digits.
3. **NullSelectionException:** Thrown when the user forgets to select a Book Category from the dropdown menu or doesn't select a Book Type (New/Old Edition).
4. **InvalidDateException:** Thrown when the date is not in the correct dd/MM/yyyy format, or if the Return Date is logically earlier than the Issue Date.

Built-in Java Exceptions:

5. **NumberFormatException:** Handled when the system tries to convert string inputs (like dates or roll numbers) into integers using Integer.parseInt(), and the input contains invalid characters.
6. **Exception (General Catch-all):** A generic exception handler added at the end to safely catch any other unexpected run-time errors that might crash the program.

Screenshots:

Library Book Issue System

Library Book Issue System

Student Name	
Roll Number	L1F23BSSE0336
Book Title	Operating Systems
Book Category	
Book Type	
Issue Date	
Return Date	7-24-2026
Remarks	Okay

Error

 Please fill all required fields!

Library Book Issue System

Library Book Issue System

Student Name

Farrukh Shahid

Roll Number

ABCD

Book Title

DSA

Book Category

Book Type

Issue Date

Return Date

25-05-2026

Remarks

Okay

Issue Book

Reset

Exit

Error

Roll Number should contain only digits!

OK

Library Book Issue System

Library Book Issue System

Student Name

Farrukh

Roll Number

0336

Book Title

DSA

Book Category

Error

Book Type

Issue Date

Return Date

27-05-2026

Remarks

Okay

Please select a Book Category!

OK

Issue Book

Reset

Exit

Library Book Issue System

Library Book Issue System

Student Name

Farrukh

Roll Number

0336

Book Title

DSA

Book Category

Book Type

Issue Date

Return Date

21-05-2026

Remarks

Okay

Error

 Date format should be dd/MM/yyyy!

OK

Issue Book

Reset

Exit

Library Book Issue System

Library Book Issue System

Student Name	Farrukh
Roll Number	0336
Book Title	DSA
Book Category	Info
Book Type	
Issue Date	
Return Date	27-05-2026
Remarks	Okay

Issue Book

Reset

Exit

Operation Completed

Library Book Issue System

Library Book Issue System

Issue and manage books with ease

Student Name

Farrukh Shahid

Roll Number

L1F23BSSE0336

Book Title

Book Category

Book Type

Issue Date

Return Date

Remarks
(optional)

Good Experience.

Success

===== Book Issued Successfully! =====

Student Name: Farrukh Shahid
Roll Number: L1F23BSSE0336
Book Title: Operating Systems
Category: Programming
Book Type: New Edition
Issue Date: 05-13-2026
Return Date: 05-14-2027
Remarks: Good Experience.

OK

Issue Book

Reset

Exit